GNU Emacs

The Emacs Speaks Statistics (ESS) package provides support to work with statistical packages. It also has support to work with R code and GNU Emacs. You can add the following to your Cask file, and install the ESS package:

```
(depends-on "ess")
```

You can add the following to your GNU Emacs initialisation file in order to use the R mode support:

```
(require 'ess-r-mode)
```

The *M-x ess-version* command provides the installed version number, which is 20211204.958 from the ELPA repository on GNU Emacs 27.2. You can start an R session from Emacs using *M-x R*. Emacs will prompt for a starting working directory, and a new buffer with an R prompt will be displayed, as shown below:

```
R version 4.1.0 (2021-05-18) -- "Camp Pontanezen"
Copyright (C) 2021 The R Foundation for Statistical Computing
Platform: x86_64-pc-linux-gnu (64-bit)
R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

  Natural language support but running in an English locale
R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.
Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.
> setwd('/home/guest/working/')
>
```

A number of commands have been implemented to interact between R code and the ESS process. A summary of the available commands to evaluate code is given in Table 1.

The advantage of using an R session within GNU Emacs is that you can save the session for future reference. You can also use ESS with the TRAMP (transparent remote access, multiple protocols) package to start a remote session. The same can be initiated with the *ess-remote* command. Do read the 'ESS - Emacs Speaks Statistics' PDF manual to learn more about its use.

The Babel project provides the ability to execute R code in an Emacs org file. The following code snippet enables support for R Babel and needs to be included in the Emacs initialisation file:

Command	Comments
ess-eval-region-or-line-and-step	Evaluate highlighted region or current line and step to next line of code
ess-eval-region-or-function-or-paragragh	Send selected region or function or paragraph
ess-eval-line	Send current line
ess-eval-line-and-go	Send line and switch point to the ESS process
ess-eval-function	Send function
ess-eval-function-and-go	Send function and switch point to the ESS process
ess-eval-buffer	Send the current buffer
ess-eval-buffer-and-go	Send current buffer and switch point to the ESS process

Table 1

```
(org-babel-do-load-languages
 'org-babel-load-languages
 '((R . t)))
```

You can enclose R code within the org Babel code blocks and execute the same in an Emacs org file to obtain the results as shown below:

```
#+BEGIN_SRC R
  1 < 2
  #+END_SRC

  #+RESULTS:
  : TRUE
```

RStudio

RStudio is an integrated development environment (IDE) for the R programming language. It can be used as a desktop application, as well as a server that can be accessed on a browser. It is primarily written in Java, C++ and JavaScript, and the open source edition is available at *https://github.com/rstudio/rstudio*. Refer to Figure 1 for RStudio.

Shiny

Shiny is an R package that is useful to develop interactive Web applications with R. You can install it with your R environment using:

```
> install.packages("shiny")
  ...
  * installing *source* package 'shiny' ...
  ** package 'shiny' successfully unpacked and MD5 sums
checked
  ** using staged installation
```